Ethical Framework for Invisible Groups
A Corrected Information-Theoretic Model

Based on the Adoptee Test and Identity Integrity

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Abstract

Many ethical scoring systems assign near-zero weight to invisible groups, allowing severe harm to go undetected. We present a corrected mathematical framework that unifies harm, visibility, identity continuity, and information loss. The Adoptee Test uses surprisal (negative log detection probability) to trigger an alarm when high-harm groups have low visibility. Boundary conditions are treated as discrete jumps (not derivatives). Identity entropy measures truth lost about origins. The Ethical Validity Ratio is shown to be recall (sensitivity). A worked example and real-world mapping demonstrate the framework's applicability.

1 Notation Reconciliation

Old symbol	**Problem**	**New symbol (v3)**
$H(g)$ (harm)	Collision with entropy H	$\mathcal{H}(g)$
H (entropy)	Collision	H_{id} (identity entropy)
$W(g)$	Overload (weight, visibility, detection)	$w(g) \in (0, 1]$ (detection probability)
T, L	Declared but unused	Removed; replaced by ΔH_{id}

2 Weighted Ethical Score

$$E(S) = \sum_{g \in G} w(g)\big(R(g) - \mathcal{H}(g)\big)$$

If $w(g) \to 0$, the group's suffering does not affect $E(S)$ — the core problem.

3 Adoptee Test – Omission Alarm (Surprisal Form)

$$A(g) = \mathcal{H}(g) \cdot \big(-\log w(g)\big)$$

where $-\log w(g)$ is the information content (surprisal) of detecting the group. When $w(g) = 1$, $A(g) = 0$. As $w(g) \to 0$, $-\log w(g) \to \infty$, so $A(g) \to \infty$.

Interpretation: High harm + invisibility → the ethical framework itself is defective.

4 Boundary-Condition Test (Discrete, Multiplicative Identity)

Define Identity Integrity $I = F \cdot K \cdot C \cdot N$, where:

- F = factual knowledge of origins
- K = kinship continuity
- C = cultural identity
- N = legal identity continuity

Each factor $\in [0, 1]$. Multiplicative: if any factor $\to 0$, then $I \to 0$ (additive would mask severance).

Adoption is a discrete event. Let $\Delta I = I_{\text{after}} - I_{\text{before}}$. The framework must detect $|\Delta I| > \tau$ for a threshold τ. If not, Framework Failure = 1. (We replace the original dI/dt, which is undefined at a step.)

5 Information-Theoretic Formulation

Ground truth about a person's origin is a point mass → true entropy $H^*_{\text{id}} = 0$. Identity entropy:

$$H_{\text{id}} = -\sum p(x) \log p(x)$$

Truth lost = increase in entropy: $L = \Delta H_{\text{id}} = H_{\text{id}} - 0 = H_{\text{id}}$. But $H_{\text{id}} = -\log w(g)$ for the group g because $w(g)$ is the probability the framework retains the truth. Thus $\Delta H_{\text{id}} = -\log w(g)$, unifying Sections 2, 4, and 5.

6 Ethical Validity Ratio as Recall

Let TP = rights protected, FN = rights ignored. Then:

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN} = \frac{\text{Rights Protected}}{\text{Rights Protected} + \text{Rights Ignored}}$$

Naming it recall allows adding precision and F_1-score. When hidden rights (e.g., knowledge of origins) are added to FN, recall collapses.

7 Worked Example: Adoptee Group

Let a group of adoptees have:

- Harm $\mathcal{H}(g) = 100$ (e.g., standardized measure from loss of origins, legal discontinuity)
- Detection probability $w(g) = 0.1$ (10% visibility in the ethical framework)

Then:
$$A(g) = 100 \times (-\log 0.1) = 100 \times \log(10) \approx 100 \times 2.3026 \approx 230.26$$
The alarm is high, indicating a severe defect in the framework. The Weighted Ethical Score contribution before correction was $w(g)(R - H) = 0.1 \times (R - 100)$. Even if $R = 0$, this contributes only -10 to $E(S)$, easily outweighed by visible groups.

8 Real-World Mapping: Sealed Birth Records

Consider a US state that seals original birth records for adopted adults. The ethical framework used by the state's review board might measure:

- Rights protected: access to medical history (partial)
- Rights ignored: knowledge of identity, heritage, biological family

Detection probability $w(g)$ for adoptees is low because the board does not include adoptee representatives. Using our framework, the omission alarm triggers, and recall collapses when FN (ignored rights) are counted. This predicts that the state's self-assessment will falsely claim high ethical validity.

9 Conclusion

The corrected framework unifies harm, visibility, entropy, and recall into a single information-theoretic argument. The Adoptee Test acts as a canary: if a system cannot detect harm to invisible groups, the system itself is unethical. By fixing notation collisions, replacing derivatives with discrete jumps, and naming recall, the framework becomes mathematically coherent and immune to technical dismissal.

Legend (Colors)

Green: protection / continuity / retained rights

Red: harm / loss / ignored rights

Blue: visibility / detection / evaluation ($w(g)$)

Orange: knowledge / origins (F)

Purple: legal / identity / power (N, P)